

Johan Lakshmanan

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EDUCATION

University of Massachusetts, Amherst

Amherst, MA

B.S. in Computer Science — GPA: 3.6/4.0, Dean's List

Aug. 2023 – May 2026

- Relevant Coursework: Operating Systems, Distributed Systems, Computer Networks, Algorithms, ML

EXPERIENCE

Roblox

May 2026 – Aug. 2026

Infrastructure Engineering Intern

San Mateo, CA

- Designed and deployed a scalable delivery architecture (versioned builds, CI/CD, first-boot verification) for **2,400+** bare-metal nodes, underpinning a **\$20M** cloud-to-on-prem GPU repatriation effort.
- Engineered backward-compatible health checks and dual-mode monitoring, enabling a zero-downtime, incremental deployment across **10** production clusters and improving node readiness by **~60%**.
- Operated as pipeline DRI supporting ML infrastructure for **70M+** daily users: merged **36+** PRs, shipped production traffic on day 24, and eliminated a class of SEV0 incidents that previously caused **5+** hours of ML model degradation.

Fidelity Investments

Jun. 2025 – Aug. 2025

Quantitative Research Intern

Boston, MA

- Piloted a model-driven trade router, scaling trade volume by **1.4x** and saving **\$200K+** over **300+** hedge funds.
- Developed and integrated backend machine learning models in Python and Kdb+/Q to optimize routing for **25M+** trades, improving selection accuracy by **20%** and reducing decision latency below **10 ms**.
- Engineered scalable ML infrastructure on AWS EKS using Jenkins CI/CD with containerized microservices and MLflow, supporting reproducible model versioning and rollback in production.

Waters Corporation

May 2024 – Aug. 2024

Software Engineering Intern

Milford, MA

- Designed multiple REST APIs to retrieve customer SharePoint resources by environment ID, decreasing mitigation times from **hours to minutes** for runtime incidents, reducing loads for on-call engineering teams.
- Developed high-concurrency C# modules to interface with SAP S/4HANA, optimizing SQL query execution and reducing transaction latency by **25%**.

PROJECTS

UMass Consulting Group Website | MERN, JavaScript, REST APIs

Feb. 2024 – May 2026

- Led full-stack development of a consulting platform using MERN stack (MongoDB, Express, React, Node.js).
- Rebuilt RESTful APIs with pagination, Redis caching, and profiling, reducing latency by **40%** and supporting **5x** higher concurrent request volumes.
- Deployed MERN platform on Dockerized AWS EC2 with Nginx load balancing and GitHub Actions CI/CD.

AI Bitcoin Sentiment Platform | Python, FastAPI, SQLAlchemy, NLP, Docker

Aug. 2025 – Jun. 2026

- Built a modular ETL pipeline aggregating cryptocurrency news, Reddit, and price feeds.
- Designed and deployed a FastAPI microservice for scheduled ingestion, FinBERT-based sentiment inference, and caching, optimizing inference latency by **120ms** and ensuring **95%+** uptime for scheduled data ingestion.

SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL, HTML/CSS, C#, Kdb+/Q

Infrastructure & DevOps: AWS (EC2, EKS), Docker, Kubernetes, CI/CD Pipelines, Terraform, Jenkins, GHA, Nginx

Frameworks: React, Node.js, Express, Django, Redis, FastAPI, Pandas, Material-UI, RESTful APIs

Systems & Tools: Git, Linux, Distributed Systems, Networking, Microservices, Packer, VFIO, Bash, MIG, CUDA

AWARDS

Hack UMass — **Best ML Hack**

Nov. 2025

Collegiate Penetration Testing Competition — **3rd Place (Global Finals)**

Jan. 2024